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$$I = \int_1^5 \frac{x^7}{(6-x)^7 + x^7} dx$$

$$\text{Stel } x = 6 - t \Rightarrow dx = -dt$$

Grenzen:

$$x = 1 \Rightarrow t = 5$$

$$x = 5 \Rightarrow t = 1$$

$$\begin{aligned} I &= \int_5^1 \frac{(6-t)^7}{t^7 + (6-t)^7} (-dt) \\ &= \int_1^5 \frac{(6-x)^7}{x^7 + (6-x)^7} dx \\ I + I &= \int_1^5 \frac{x^7 + (6-x)^7}{x^7 + (6-x)^7} dx \\ 2I &= \int_1^5 1 dx \\ &= [x]_1^5 = 5 - 1 = 4 \\ I &= 2 \end{aligned}$$

References